
HPC Cluster Project Documentation

Course: CNT316D

Project: Creating a 3-Node HPC Cluster and Running Benchmarks

Group name: Ignite Network Engineers

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Student Number: 223719481

Date: 1 April 2026

ATTENDANCE

1. Summary

Meeting title	Ignite Network Engineers
Attended participants	4
Start time	4/01/26, 6:09:10 PM
End time	4/01/26, 8:22:32 PM
Meeting duration	2h 13m 22s
Average attendance time	2h 8m 14s

2. Participants

Name	First Join	Last Leave	In-Meeting Duration
J Sefaan	4/01/26, 6:09:12 PM	4/01/26, 8:22:32 PM	2h 13m 19s
Olwethu Hlabisa	4/01/26, 6:10:07 PM	4/01/26, 8:20:12 PM	2h 10m 4s
S Molaba	4/01/26, 6:12:15 PM	4/01/26, 8:16:45 PM	2h 4m 29s
D Tshabangu	4/01/26, 6:16:37 PM	4/01/26, 8:21:41 PM	2h 5m 3s
3. In-Meeting Activities			
Name	Join Time	Leave Time	Duration
J Sefaan	4/01/26, 6:09:12 PM	4/01/26, 8:22:32 PM	2h 13m 19s
Olwethu Hlabisa	4/01/26, 6:10:07 PM	4/01/26, 8:20:12 PM	2h 10m 4s
S Molaba	4/01/26, 6:12:15 PM	4/01/26, 8:16:45 PM	2h 4m 29s
D Tshabangu	4/01/26, 6:16:37 PM	4/01/26, 8:21:41 PM	2h 5m 3s

Link for recording: [Recap: Ignite Network Engineers Monday, March 30 | Meeting | Microsoft Teams](#)

1. Introduction

- This project focuses on designing and implementing a 3-node High Performance Computing (HPC) cluster using Rocky Linux and Slurm workload manager. The aim is to simulate a distributed computing environment and evaluate system performance using industry-standard benchmarks such as HPL, LAMMPS, HPCC, PennyLane, and MILC.
- WHAT IS AN HPC CLUSTER?

A High-Performance Computing (HPC) cluster is a group of interconnected computers (nodes) that work together as a single system to perform complex computations. It consists of:

- Head node (controls jobs)
- Compute nodes (execute tasks)

- Shared storage
- Network communication

➤ SYSTEM ARCHITECTURE

The system consists of three nodes:

- Head Node: Responsible for job scheduling and management
- Compute Node 1 & 2: Execute parallel tasks

➤ TECHNOLOGIES USED

Slurm

- Slurm (Simple Linux Utility for Resource Management) is a workload manager used to schedule and manage jobs in an HPC cluster. It allocates resources such as CPU and memory to tasks and ensures efficient execution.

MPI (OpenMPI)

MPI (Message Passing Interface) is used for communication between nodes in a cluster. It allows processes running on different machines to exchange data during execution.

NFS

NFS (Network File System) enables shared storage across all nodes, allowing them to access the same files and data.

SSH

SSH is used to enable secure communication between nodes. Passwordless SSH is configured to allow automated job execution.

MUNGE

MUNGE is an authentication service used in HPC clusters to validate communication between nodes in Slurm.

➤ BENCHMARKS

1. HPL (High Performance Linpack)

HPL is a benchmark used to measure the floating-point computing power of a system by solving linear equations.

- Purpose: Measure CPU performance
 - Expected Result: Higher FLOPS = better performance
-

2. LAMMPS

LAMMPS is a molecular dynamics simulator used for modeling particles and atomic interactions.

- Purpose: Test parallel processing
- Expected Result: Faster simulation with more nodes

3. HPCC

HPCC is a benchmark suite that evaluates multiple aspects such as memory bandwidth, network speed, and CPU performance.

- Purpose: Full system performance test
 - Expected Result: Balanced system performance
-

4. PennyLane

PennyLane is a quantum computing framework used for hybrid quantum-classical computations.

- Purpose: Evaluate advanced computational workloads
- Expected Result: Efficient execution of quantum simulations

5. MILC

MILC is used for quantum chromodynamics simulations in physics.

- Purpose: Test parallel scientific computing
- Expected Result: Efficient scaling across nodes

6. MATLAB Optimization

MATLAB scripts are optimized using parallel computing techniques to improve execution time.

- Purpose: Improve performance
- Expected Result: Reduced runtime

7. EXPECTED VS ACTUAL RESULTS

The expected outcome is that performance improves as more nodes are used. However, network latency and system limitations may affect actual performance.

Reality check:

- **Network latency** between nodes can slightly reduce speedup.
- **Compute node limitations** (CPU cores, memory) can prevent linear scaling.
- **Storage bottlenecks** if the shared NFS directory is heavily accessed.

8. SCALING

Scaling shows how well the cluster handles larger workloads or more nodes.

Strong Scaling

Same problem, more nodes, faster execution

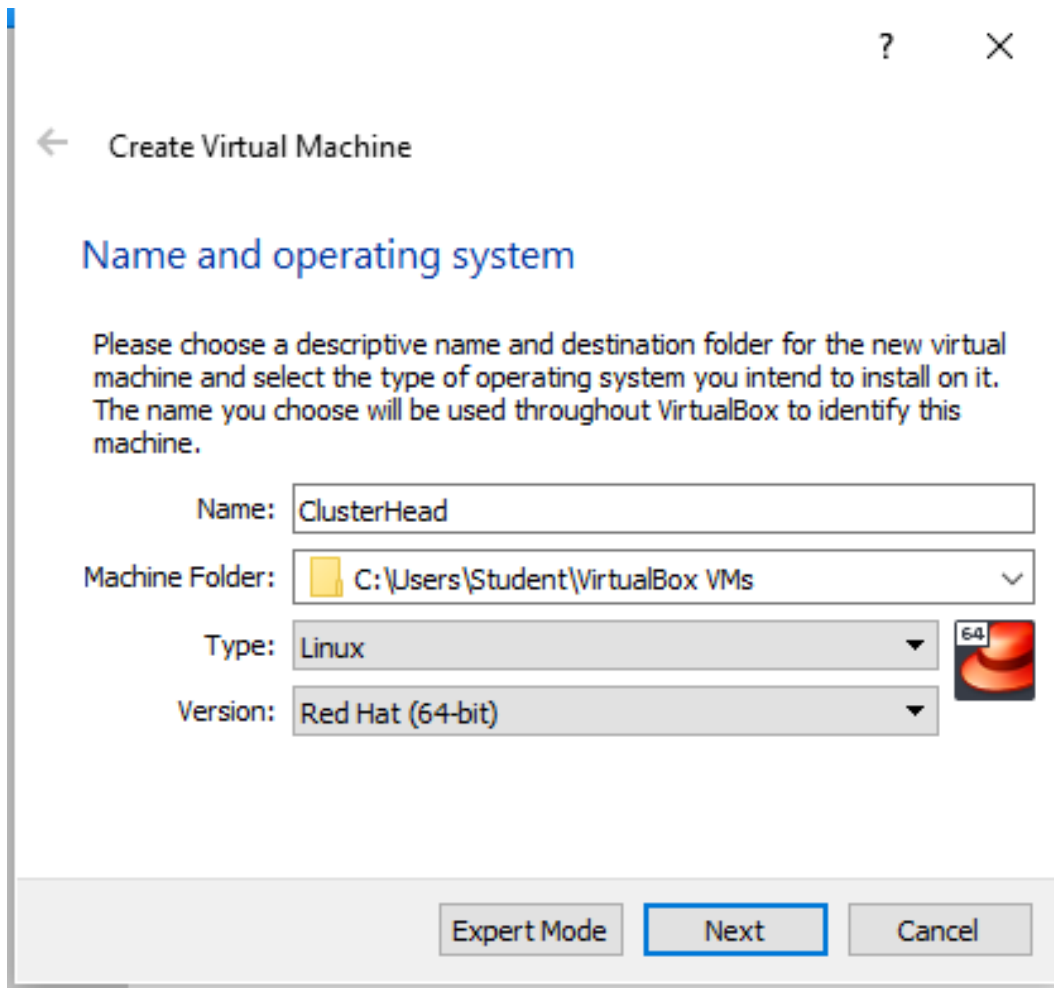
Weak Scaling

Bigger problem, more nodes, same execution time

9. SECURITY

The cluster uses SSH key-based authentication to ensure secure communication. Firewall and SELinux settings are configured to allow cluster communication while maintaining security.

2. Virtual Machine Creation



← Create Virtual Machine

Name and operating system

Please choose a descriptive name and destination folder for the new virtual machine and select the type of operating system you intend to install on it. The name you choose will be used throughout VirtualBox to identify this machine.

Name: ClusterHead

Machine Folder: C:\Users\Student\VirtualBox VMs

Type: Linux

Version: Red Hat (64-bit)

Expert Mode Next Cancel

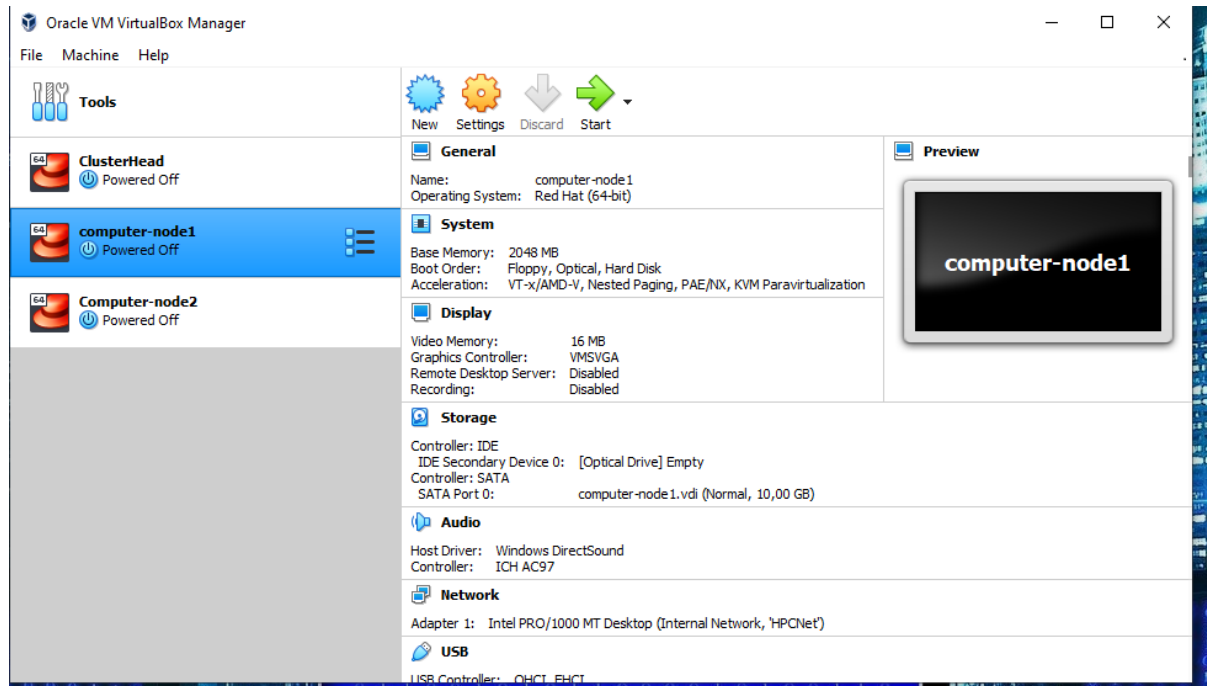
2.1 Head Node

- Name: head-node
- Type: Linux
- Version: Red Hat (64-bit)
- RAM: 2048 MB
- CPU: 2 cores
- Disk: 20 GB
- Network: Host-Only Adapter

2.2 Compute Node 1

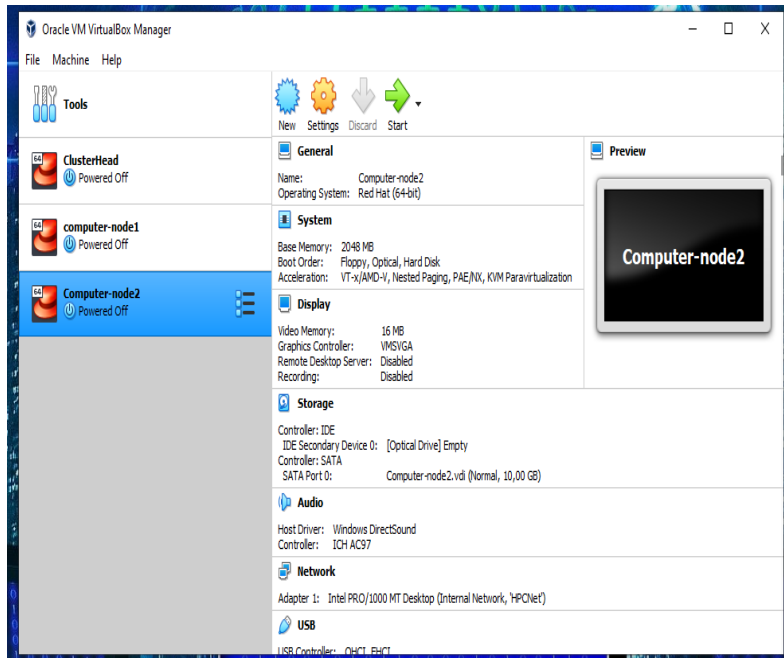
- Name: compute-node-1
- Type: Linux

- Version: Red Hat (64-bit)
- RAM: 2048 MB
- CPU: 1–2 cores
- Disk: 10 GB
- Network: Host-Only Adapter



2.3 Compute Node 2

- Name: compute-node-2
- Same specifications as Compute Node 1



3. Installation of Rocky Linux

- Minimal installation
- Root password and user 'hpcuser' created
- Hostnames set

Commands

```
sudo dnf update -y
```

```
sudo hostname
```

Screenshot placeholder: Terminal showing update

Head node:


```

Installed:
PackageKit-1.1.12-7.el8.x86_64
bind-libs-lite-32:9.11.36-16.el8_10.6.x86_64
cairo-1.15.12-6.el8.x86_64
cockpit-packagekit-310.3-3.el8_10.noarch
gdk-pixbuf2-2.36.12-7.el8_10.x86_64
grub2-tools-efi-1:2.02-170.el8_10.1.rocky.0.1.x86_64
kernel-modules-4.18.0-553.111.1.el8_10.x86_64
libXau-1.0.9-3.el8.x86_64
libappstream-glib-0.7.14-3.el8.x86_64
libsecret-0.18.6-1.el8_0.2.x86_64
libxcb-1.13.1-1.el8.x86_64
pixman-0.38.4-4.el8.x86_64
python3-audit-3.1.2-1.el8_10.1.x86_64
python3-distro-1.4.0-2.module+el8.10.0+1910+234ad790.noarch
python3-libxml2-2.9.7-21.el8_10.3.x86_64
python3-policycoreutils-2.9-26.el8_10.noarch
python3-pydbus-0.6.0-5.el8.noarch
python3-unbound-1.16.2-5.9.el8.x86_64
sscg-3.0.0-7.el8.x86_64
xkeyboard-config-2.28-1.el8.noarch

PackageKit-glib-1.1.12-7.el8.x86_64
bind-license-32:9.11.36-16.el8_10.6.noarch
cairo-gobject-1.15.12-6.el8.x86_64
fontconfig-2.13.1-4.el8.x86_64
geolite2-city-20180605-1.el8.noarch
kernel-4.18.0-553.111.1.el8_10.x86_64
libX11-1.6.8-9.el8_10.x86_64
libXext-1.3.4-1.el8.x86_64
libevent-2.1.8-5.el8.x86_64
libsoup-2.62.3-13.el8_10.x86_64
libxkbcommon-0.9.1-1.el8.x86_64
policycoreutils-python-utils-2.9-26.el8_10.noarch
python3-bind-32:9.11.36-16.el8_10.6.noarch
python3-gobject-3.28.3-2.el8.x86_64
python3-ipect-4.3.1-3.el8.noarch
python3-psutil-5.4.3-11.el8.x86_64
python3-setools-4.3.0-5.el8.x86_64
setroubleshoot-plugins-3.3.14-1.el8.noarch
tracer-common-1.1-1.el8.noarch

bind-libs-32:9.11.36-16.el8_10.6.x86_64
bind-utils-32:9.11.36-16.el8_10.6.x86_64
checkpolicy-2.9-1.el8.x86_64
fstirm-0.6.1-3.el8.x86_64
geolite2-country-20180605-1.el8.noarch
kernel-core-4.18.0-553.111.1.el8_10.x86_64
libX11-common-1.6.8-9.el8_10.noarch
libXrender-0.9.10-7.el8.x86_64
libmaxminddb-1.2.0-10.el8_9.1.x86_64
libstemmer-0-10.585svn.el8.x86_64
pinentry-1.1.0-2.el8.x86_64
protobuf-c-1.3.0-8.el8.x86_64
python3-cairo-1.16.3-6.el8.x86_64
python3-libsemanage-2.9-12.el8_10.x86_64
python3-ply-3.9-9.el8.noarch
python3-ptyprocess-0.5.2-4.el8.noarch
python3-tracer-1.1-1.el8.noarch
setroubleshoot-server-3.3.26-6.el8.x86_64
unbound-libs-1.16.2-5.9.el8.x86_64

Complete!
ignitecompute@headnode ~]$

```

Compute-node1:

```

Installed:
PackageKit-1.1.12-7.el8.x86_64
bind-libs-lite-32:9.11.36-16.el8_10.6.x86_64
cairo-1.15.12-6.el8.x86_64
cockpit-packagekit-310.3-3.el8_10.noarch
gdk-pixbuf2-2.36.12-7.el8_10.x86_64
grub2-tools-efi-1:2.02-170.el8_10.1.rocky.0.1.x86_64
kernel-modules-4.18.0-553.111.1.el8_10.x86_64
libXau-1.0.9-3.el8.x86_64
libappstream-glib-0.7.14-3.el8.x86_64
libsecret-0.18.6-1.el8_0.2.x86_64
libxcb-1.13.1-1.el8.x86_64
pixman-0.38.4-4.el8.x86_64
python3-audit-3.1.2-1.el8_10.1.x86_64
python3-distro-1.4.0-2.module+el8.10.0+1910+234ad790.noarch
python3-libxml2-2.9.7-21.el8_10.3.x86_64
python3-policycoreutils-2.9-26.el8_10.noarch
python3-pydbus-0.6.0-5.el8.noarch
python3-unbound-1.16.2-5.9.el8.x86_64
sscg-3.0.0-7.el8.x86_64
xkeyboard-config-2.28-1.el8.noarch

PackageKit-glib-1.1.12-7.el8.x86_64
bind-license-32:9.11.36-16.el8_10.6.noarch
cairo-gobject-1.15.12-6.el8.x86_64
fontconfig-2.13.1-4.el8.x86_64
geolite2-city-20180605-1.el8.noarch
kernel-4.18.0-553.111.1.el8_10.x86_64
libX11-1.6.8-9.el8_10.x86_64
libXext-1.3.4-1.el8.x86_64
libevent-2.1.8-5.el8.x86_64
libsoup-2.62.3-13.el8_10.x86_64
libxkbcommon-0.9.1-1.el8.x86_64
policycoreutils-python-utils-2.9-26.el8_10.noarch
python3-bind-32:9.11.36-16.el8_10.6.noarch
python3-gobject-3.28.3-2.el8.x86_64
python3-ipect-4.3.1-3.el8.noarch
python3-psutil-5.4.3-11.el8.x86_64
python3-setools-4.3.0-5.el8.x86_64
setroubleshoot-plugins-3.3.14-1.el8.noarch
tracer-common-1.1-1.el8.noarch

bind-libs-32:9.11.36-16.el8_10.6.x86_64
bind-utils-32:9.11.36-16.el8_10.6.x86_64
checkpolicy-2.9-1.el8.x86_64
fstirm-0.6.1-3.el8.x86_64
geolite2-country-20180605-1.el8.noarch
kernel-core-4.18.0-553.111.1.el8_10.x86_64
libX11-common-1.6.8-9.el8_10.noarch
libXrender-0.9.10-7.el8.x86_64
libmaxminddb-1.2.0-10.el8_9.1.x86_64
libstemmer-0-10.585svn.el8.x86_64
pinentry-1.1.0-2.el8.x86_64
protobuf-c-1.3.0-8.el8.x86_64
python3-cairo-1.16.3-6.el8.x86_64
python3-libsemanage-2.9-12.el8_10.x86_64
python3-ply-3.9-9.el8.noarch
python3-ptyprocess-0.5.2-4.el8.noarch
python3-tracer-1.1-1.el8.noarch
setroubleshoot-server-3.3.26-6.el8.x86_64
unbound-libs-1.16.2-5.9.el8.x86_64

Complete!
lcompute-node1@compute-node1 ~]$
lcompute-node1@compute-node1 ~]$
lcompute-node1@compute-node1 ~]$

```

Compute-node2:

```

Installed:
PackageKit-1.1.12-7.el8.x86_64
bind-libs-lite-32:9.11.36-16.el8_10.6.x86_64
cairo-1.15.12-6.el8.x86_64
cockpit-packagekit-318.3-3.el8_10.noarch
gdk-pixbuf2-2.36.12-7.el8_10.x86_64
grub2-tools-efi-1:2.02-170.el8_10.1.rocky.0.1.x86_64
kernel-modules-4.18.0-553.111.1.el8_10.x86_64
libXau-1.0.9-3.el8.x86_64
libappstream-glib-0.7.14-3.el8.x86_64
libsecret-0.18.6-1.el8_0.2.x86_64
libxch-1.13.1-1.el8.x86_64
pixman-0.38.4-4.el8.x86_64
python3-audit-3.1.2-1.el8_10.1.x86_64
python3-distro-1.4.0-2.module+el8.10.0+1918+234ad790.noarch
python3-libxml2-2.9.7-21.el8_10.3.x86_64
python3-policycoreutils-2.9-26.el8_10.noarch
python3-pydbus-0.6.0-5.el8.noarch
python3-unbound-1.16.2-5.9.el8.x86_64
sscg-3.0.0-7.el8.x86_64
xkeyboard-config-2.28-1.el8.noarch

PackageKit-glib-1.1.12-7.el8.x86_64
bind-license-32:9.11.36-16.el8_10.6.noarch
cairo-gobject-1.15.12-6.el8.x86_64
fontconfig-2.13.1-4.el8.x86_64
geolite2-city-20180605-1.el8.noarch
kernel-4.18.0-553.111.1.el8_10.x86_64
libX11-1.6.8-9.el8_10.x86_64
libXext-1.3.4-1.el8.x86_64
libevent-2.1.8-5.el8.x86_64
libsoup-2.62.3-13.el8_10.x86_64
libxkbcommon-0.9.1-1.el8.x86_64
policycoreutils-python-utils-2.9-26.el8_10.noarch
python3-bind-32:9.11.36-16.el8_10.6.noarch
python3-gobject-3.28.3-2.el8.x86_64
python3-pexpect-4.3.1-3.el8.noarch
python3-psutil-5.4.3-11.el8.x86_64
python3-setools-4.3.0-5.el8.x86_64
setroubleshoot-plugins-3.3.14-1.el8.noarch
tracer-common-1.1-1.el8.noarch

bind-libs-32:9.11.36-16.el8_10.6.x86_64
bind-utils-32:9.11.36-16.el8_10.6.x86_64
checkpolicy-2.9-1.el8.x86_64
fstirm-0.6.1-3.el8.x86_64
geolite2-country-20180605-1.el8.noarch
kernel-core-4.18.0-553.111.1.el8_10.x86_64
libX11-common-1.6.8-9.el8_10.noarch
libXrender-0.9.10-7.el8.x86_64
libmaxminddb-1.2.0-10.el8_9.1.x86_64
libstemmer-0-10.585son.el8.x86_64
pinentry-1.1.0-2.el8.x86_64
protobuf-c-1.3.0-8.el8.x86_64
python3-cairo-1.16.3-6.el8.x86_64
python3-libsemanage-2.9-12.el8_10.x86_64
python3-ply-3.9-9.el8.noarch
python3-ptyprocess-0.5.2-4.el8.noarch
python3-tracer-1.1-1.el8.noarch
setroubleshoot-server-3.3.26-6.el8.x86_64
unbound-libs-1.16.2-5.9.el8.x86_64

Complete!
[computer-node2@computernode2 ~]$

```

Host names:

sudo hostname head-node

```

[lignitecompute@headnode ~]$ sudo hostname head-node
[sudo] password for ignitecompute:
[lignitecompute@headnode ~]$ hostname
head-node
[lignitecompute@headnode ~]$

```

sudo hostname compute-node-1

```

[computer-node1@computernode1 ~]$ sudo hostname compute-node-1
[computer-node1@computernode1 ~]$ hostname
compute-node-1
[computer-node1@computernode1 ~]$

```

sudo hostname compute-node-2

```

[computer-node2@computernode2 ~]$ sudo hostname compute-node-2
[sudo] password for computer-node2:
[computer-node2@computernode2 ~]$ hostname
compute-node-2
[computer-node2@computernode2 ~]$

```

4. Network Configuration (Static IP)

Head Node

ip addr

sudo nmcli con mod " enp0s8" ipv4.addresses 192.168.231.10/24 ipv4.gateway 192.168.231.1 ipv4.dns 8.8.8.8 ipv4.method manual

```

ignitecompute@headnode ~$ ip route
default via 10.58.110.17 dev enp0s3 proto dhcp src 10.58.110.102 metric 100
10.58.110.0/24 dev enp0s3 proto kernel scope link src 10.58.110.102 metric 100
192.168.231.0/24 dev enp0s8 proto kernel scope link src 192.168.231.3 metric 101
ignitecompute@headnode ~$ nmcli con show
NAME UUID TYPE DEVICE
enp0s3 b50053a1-f43a-4dee-9822-d85a1c119a3a ethernet enp0s3
enp0s8 63e74277-07a2-4ef7-ae89-5a5e2264e952 ethernet enp0s8
ignitecompute@headnode ~$ sudo nmcli con mod "enp0s8" ipv4.addresses 192.168.231.10/24 ipv4.gateway 192.168.231.1 ipv4.dns "8.8.8.8" ipv4.method manual
[sudo] password for ignitecompute:
Sorry, try again.
[sudo] password for ignitecompute:
ignitecompute@headnode ~$ sudo nmcli con down "enp0s8"
Connection 'enp0s8' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
ignitecompute@headnode ~$ sudo nmcli con up "enp0s8"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
ignitecompute@headnode ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f9:3b:03 brd ff:ff:ff:ff:ff:ff
    inet 10.58.110.102/24 brd 10.58.110.255 scope global dynamic noprefixroute enp0s3
        valid_lft 1972sec preferred_lft 1972sec
    inet6 fe80::a00:27ff:fe9:3b03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f3:8b:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.231.10/24 brd 192.168.231.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::236a:3d72:fb70:eb2a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ignitecompute@headnode ~$ ip route
default via 10.58.110.17 dev enp0s3 proto dhcp src 10.58.110.102 metric 100
default via 192.168.231.1 dev enp0s8 proto static metric 101
10.58.110.0/24 dev enp0s3 proto kernel scope link src 10.58.110.102 metric 100
192.168.231.0/24 dev enp0s8 proto kernel scope link src 192.168.231.10 metric 101
ignitecompute@headnode ~$

```

Compute-node-1

ip addr

```
sudo nmcli con mod " enp0s8" ipv4.addresses 192.168.231.11/24 ipv4.gateway 19
2.168.231.1 ipv4.dns 8.8.8.8 ipv4.method manual
```

```

[computer-node1@computer-node1 ~]$ sudo nmcli con mod "enp0s8" ipv4.addresses 192.168.231.11/24 ipv4.gateway 192.168.231.1 ipv4.dns "8.8.8.8" ipv4.method manual
[computer-node1@computer-node1 ~]$ sudo nmcli con down "enp0s8"
Connection 'enp0s8' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
[computer-node1@computer-node1 ~]$ sudo nmcli con up "enp0s8"
Error: argument 'upenp0s8' not understood. Try passing --help instead.
[computer-node1@computer-node1 ~]$ sudo nmcli con up "enp0s8"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[computer-node1@computer-node1 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:56:c6:ca brd ff:ff:ff:ff:ff:ff
    inet 10.58.110.21/24 brd 10.58.110.255 scope global dynamic noprefixroute enp0s3
        valid_lft 2457sec preferred_lft 2457sec
    inet6 fe80::a00:27ff:fe56:c6ca/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a6:94:60 brd ff:ff:ff:ff:ff:ff
    inet 192.168.231.11/24 brd 192.168.231.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::1116:591:8b61:cb6f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[computer-node1@computer-node1 ~]$

```

Compute-node-2

ip addr

```
sudo nmcli con mod " enp0s8" ipv4.addresses 192.168.231.12/24 ipv4.gateway 19
2.168.231.1 ipv4.dns 8.8.8.8 ipv4.method manual
```

```
[computer-node2@computernode2 ~]$ sudo nmcli con mod "enp0s8" ipv4.addresses 192.168.231.12/24 ipv4.gateway 192.168.231.1 ipv4.dns "8.8.8.8" ipv4.method manual
[computer-node2@computernode2 ~]$ sudo nmcli con down "enp0s8"
Connection 'enp0s8' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
[computer-node2@computernode2 ~]$ sudo nmcli con up "enp0s8"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
[computer-node2@computernode2 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:4b:4f:32 brd ff:ff:ff:ff:ff:ff
    inet 10.58.110.251/24 brd 10.58.110.255 scope global dynamic noprefixroute enp0s3
        valid_lft 3833sec preferred_lft 3833sec
    inet6 fe80::a00:27ff:fe4b:4f32/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:85:7c brd ff:ff:ff:ff:ff:ff
    inet 192.168.231.12/24 brd 192.168.231.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::e895:22bd:677e:3d2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[computer-node2@computernode2 ~]$ ip route
default via 10.58.110.17 dev enp0s3 proto dhcp src 10.58.110.251 metric 100
default via 192.168.231.1 dev enp0s8 proto static metric 101
10.58.110.0/24 dev enp0s3 proto kernel scope link src 10.58.110.251 metric 100
192.168.231.0/24 dev enp0s8 proto kernel scope link src 192.168.231.12 metric 101
[computer-node2@computernode2 ~]$
```

/etc/hosts entries

```
192.168.231.10 head-node
192.168.231.11 compute-node-1
192.168.231.12 compute-node-2
```

Screenshot placeholder: hosts file

```
GNU nano 2.9.8 /etc/hosts

127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

192.168.231.10 headnode
192.168.231.11 computernode1
192.168.231.12 computernode2
```

5. SSH Passwordless Authentication

ssh-keygen

```
ssh-copy-id ignitecompute@compute-node-1
ssh-copy-id ignitecompute@compute-node-2
ssh compute-node-1
ssh compute-node-2
```

```

ignitecompute@headnode ~]$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ignitecompute/.ssh/id_rsa):
/home/ignitecompute/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ignitecompute/.ssh/id_rsa.
Your public key has been saved in /home/ignitecompute/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:DuZ4gp/ENbeGiCsq4WKAaggEXSv0TN00UrGRgeM+u9HU ignitecompute@headnode
The key's randomart image is:
+---[RSA 4096]---+
|  ooo. oo. |
| .ooo. o.o |
| .....=..+ |
| . .+o o |
| o. =.S + |
| * o.*.=.+E. |
| +. .ooB...o |
| oo . .=. . |
| o... .o.o |
+---[SHA256]-----+
ignitecompute@headnode ~]$ ssh-copy-id computer-node1@192.168.231.11
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ignitecompute/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
computer-node1@192.168.231.11's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'computer-node1@192.168.231.11'"
and check to make sure that only the key(s) you wanted were added.

ignitecompute@headnode ~]$ ssh-copy-id computer-node2@192.168.231.12
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ignitecompute/.ssh/id_rsa.pub"
The authenticity of host '192.168.231.12 (192.168.231.12)' can't be established.
ECDSA key fingerprint is SHA256:JfCki4EuekL6XjW/waUyRtMKjDGPd62pAof96r0ZME.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
computer-node2@192.168.231.12's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'computer-node2@192.168.231.12'"
and check to make sure that only the key(s) you wanted were added.

ignitecompute@headnode ~]$

```

Successful login:

```

ignitecompute@headnode ~]$ ssh computer-node1@192.168.231.11
Enter passphrase for key '/home/ignitecompute/.ssh/id_rsa':
computer-node1@192.168.231.11's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Tue Mar 31 08:42:50 2026
[computer-node1@computernode1 ~]$

```

6. Install Required Software

```
sudo dnf install -y gcc make
```

```

Downloading Packages:
(1/9): isl-0.16.1-6.el8.x86_64.rpm                               1.0 MB/s | 834 kB   00:00
(2/9): libmpc-1.1.0-9.1.el8.x86_64.rpm                         1.9 MB/s | 60 kB   00:00
(3/9): glibc-devel-2.28-251.el8_10.31.x86_64.rpm             2.2 MB/s | 90 kB   00:00
(4/9): glibc-headers-2.28-251.el8_10.31.x86_64.rpm           3.0 MB/s | 495 kB   00:00
(5/9): cpp-8.5.0-28.el8_10.x86_64.rpm                         3.5 MB/s | 10 MB   00:02
(6/9): libxcrypt-devel-4.1.1-6.el8.x86_64.rpm                1.2 MB/s | 24 kB   00:00
(7/9): make-4.2.1-11.el8.x86_64.rpm                          3.5 MB/s | 497 kB   00:00
(8/9): kernel-headers-4.18.0-553.111.1.el8_10.x86_64.rpm     5.3 MB/s | 12 MB   00:02
(9/9): gcc-8.5.0-28.el8_10.x86_64.rpm                         4.4 MB/s | 23 MB   00:05
-----
Total                                                            7.1 MB/s | 48 MB   00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : libmpc-1.1.0-9.1.el8.x86_64                    1/1
  Installing : cpp-8.5.0-28.el8_10.x86_64                    1/9
  Running scriptlet: cpp-8.5.0-28.el8_10.x86_64                2/9
  Installing : kernel-headers-4.18.0-553.111.1.el8_10.x86_64 3/9
  Running scriptlet: glibc-headers-2.28-251.el8_10.31.x86_64 4/9
  Installing : glibc-headers-2.28-251.el8_10.31.x86_64       4/9
  Installing : libxcrypt-devel-4.1.1-6.el8.x86_64            4/9
  Installing : glibc-devel-2.28-251.el8_10.31.x86_64         5/9
  Running scriptlet: glibc-devel-2.28-251.el8_10.31.x86_64   6/9
  Installing : isl-0.16.1-6.el8.x86_64                       6/9
  Running scriptlet: isl-0.16.1-6.el8.x86_64                  7/9
  Installing : gcc-8.5.0-28.el8_10.x86_64                    7/9
  Running scriptlet: gcc-8.5.0-28.el8_10.x86_64              8/9
  Installing : make-4.2.1-11.el8.x86_64                      8/9
  Running scriptlet: make-4.2.1-11.el8.x86_64                9/9
  Verifying : cpp-8.5.0-28.el8_10.x86_64                     9/9
  Verifying : gcc-8.5.0-28.el8_10.x86_64                     1/9
  Verifying : isl-0.16.1-6.el8.x86_64                         1/9
  Verifying : libmpc-1.1.0-9.1.el8.x86_64                     2/9
  Verifying : glibc-devel-2.28-251.el8_10.31.x86_64          3/9
  Verifying : glibc-headers-2.28-251.el8_10.31.x86_64        4/9
  Verifying : kernel-headers-4.18.0-553.111.1.el8_10.x86_64 5/9
  Verifying : libxcrypt-devel-4.1.1-6.el8.x86_64             6/9
  Verifying : make-4.2.1-11.el8.x86_64                        7/9
  Verifying : gcc-8.5.0-28.el8_10.x86_64                      8/9
  Verifying : isl-0.16.1-6.el8.x86_64                         9/9
Installed:
  cpp-8.5.0-28.el8_10.x86_64    gcc-8.5.0-28.el8_10.x86_64    glibc-devel-2.28-251.el8_10.31.x86_64    glibc-headers-2.28-251.el8_10.31.x86_64
  isl-0.16.1-6.el8.x86_64      kernel-headers-4.18.0-553.111.1.el8_10.x86_64    libmpc-1.1.0-9.1.el8.x86_64    libxcrypt-devel-4.1.1-6.el8.x86_64
  make-4.2.1-11.el8.x86_64
Complete!
[ignitecompute@headnode ~]$

```

`sudo dnf install -y openmpi openmpi-devel`

```

Installed:
  environment-modules-4.5.2-4.el8.x86_64
  hwloc-libs-2.2.0-3.el8.x86_64
  libgfortran-8.5.0-28.el8_10.x86_64
  libpsm2-11.2.230-1.el8_1.x86_64
  libquadmath-devel-8.5.0-28.el8_10.x86_64
  numge-libs-0.5.13-3.el8_10.x86_64
  openmpi-devel-2.4.1.1-5.el8.x86_64
  perl-Carp-1.42-396.el8.noarch
  perl-Digest-1.17-395.el8.noarch
  perl-Encode-4.2.97-3.el8.x86_64
  perl-Exporter-5.72-396.el8.noarch
  perl-File-Temp-0.230.600-1.el8.noarch
  perl-HTTP-Tiny-0.074-3.el8.noarch
  perl-IO-Socket-IP-0.39-5.el8.noarch
  perl-MIME-Base64-3.15-396.el8.x86_64
  perl-Net-SSLeay-1.88-2.module+el8.9.0+1517+e71a7a62.x86_64
  perl-Pod-Escapes-1.1.07-395.el8.noarch
  perl-Pod-Simple-1.3.35-395.el8.noarch
  perl-Scalar-List-Utils-3.1.49-2.el8.x86_64
  perl-Storable-1.3.11-3.el8.x86_64
  perl-Term-Cap-1.17-395.el8.noarch
  perl-Text-Tabs+Wrap-2013.0523-395.el8.noarch
  perl-URI-1.73-3.el8.noarch
  perl-constant-1.33-396.el8.noarch
  perl-libnet-3.11-3.el8.noarch
  perl-macros-4.5.26.3-423.el8_10.x86_64
  perl-podlators-4.11-1.el8.noarch
  perl-threads-shared-1.58-2.el8.x86_64
  rpm-mpi-hooks-8-2.el8.noarch
  ucx-1.15.0-2.el8.x86_64
  gcc-gfortran-8.5.0-28.el8_10.x86_64
  libfabric-1.10.0-1.el8.x86_64
  libibmad-48.0-1.el8.x86_64
  libquadmath-8.5.0-28.el8_10.x86_64
  librdmacm-48.0-1.el8.x86_64
  openmpi-2.4.1.1-5.el8.x86_64
  opensm-libs-3.3.24-1.el8.x86_64
  perl-Data-Dumper-2.167-399.el8.x86_64
  perl-Digest-MD5-2.55-396.el8.x86_64
  perl-Errno-1.28-423.el8_10.x86_64
  perl-File-Path-2.15-2.el8.noarch
  perl-Getopt-Long-1.2.50-4.el8.noarch
  perl-IO-1.38-423.el8_10.x86_64
  perl-IO-Socket-SSL-2.066-4.module+el8.9.0+1517+e71a7a62.noarch
  perl-Mozilla-Ca-20160104-7.module+el8.9.0+1521+0101edce.noarch
  perl-PathTools-3.74-1.el8.x86_64
  perl-Pod-Perldoc-3.28-396.el8.noarch
  perl-Pod-Usage-4.1.69-395.el8.noarch
  perl-Socket-4.2.027-3.el8.x86_64
  perl-Term-ANSIColor-4.06-396.el8.noarch
  perl-Text-ParseWords-3.30-395.el8.noarch
  perl-Time-Local-1.1.280-1.el8.noarch
  perl-Unicode-Normalize-1.25-396.el8.x86_64
  perl-Interpreter-4.5.26.3-423.el8_10.x86_64
  perl-libs-4.5.26.3-423.el8_10.x86_64
  perl-parent-1.0.237-1.el8.noarch
  perl-threads-1.2.21-2.el8.x86_64
  pmix-2.2.5-3.el8.x86_64
  tc1-1.8.6.8-2.el8.x86_64
Complete!
[ignitecompute@headnode ~]$

```

```

(pennylane_env) [ignitecompute@headnode ~]$ module load mpi
(pennylane_env) [ignitecompute@headnode ~]$ which mpirun
/usr/lib64/openmpi/bin/mpirun
(pennylane_env) [ignitecompute@headnode ~]$ mpirun --version
mpirun (Open MPI) 4.1.1

```

Report bugs to <http://www.open-mpi.org/community/help/>

```
(pennylane_env) [ignitecompute@headnode ~]$
```

```
sudo dnf install -y slurm slurm-slurmd slurm-client
```

Enable EPEL repository, since slurm is not available in enabled repositories, the install slurm again.

```
Total 1.7 MB/s | 8.3 MB 00:04
Extra Packages for Enterprise Linux 8 - x86_64 1.6 MB/s | 1.6 kB 00:00
Importing GPG key 0x2F86D6A1:
  Userid : "Fedora EPEL (8) <epel@fedoraproject.org>"
  Fingerprint: 94E2 79EB 8D8F 25B2 1818 ADF1 21EA 45AB 2F86 D6A1
  From : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-8
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Installing : slurm-libs-20.11.9-1.el8.x86_64 1/12
  Installing : libaec-1.0.2-3.el8.x86_64 2/12
  Installing : hdf5-1.10.5-4.el8.x86_64 3/12
  Installing : libjwt-1.12.1-7.el8.x86_64 4/12
  Installing : infiniband-diags-48.0-1.el8.x86_64 5/12
  Installing : freeipmi-1.6.14-2.el8.x86_64 6/12
Running scriptlet: freeipmi-1.6.14-2.el8.x86_64 6/12
Running scriptlet: munge-0.5.13-3.el8_10.x86_64 7/12
  Installing : munge-0.5.13-3.el8_10.x86_64 7/12
Running scriptlet: munge-0.5.13-3.el8_10.x86_64 7/12
  Installing : mariadb-connector-c-config-3.1.11-2.el8_3.noarch 8/12
  Installing : mariadb-connector-c-3.1.11-2.el8_3.x86_64 9/12
  Installing : slurm-20.11.9-1.el8.x86_64 10/12
  Installing : slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
Running scriptlet: slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
  Installing : slurm-slurmd-20.11.9-1.el8.x86_64 12/12
Running scriptlet: slurm-slurmd-20.11.9-1.el8.x86_64 12/12
Verifying : mariadb-connector-c-3.1.11-2.el8_3.x86_64 1/12
Verifying : mariadb-connector-c-config-3.1.11-2.el8_3.noarch 2/12
Verifying : munge-0.5.13-3.el8_10.x86_64 3/12
Verifying : freeipmi-1.6.14-2.el8.x86_64 4/12
Verifying : infiniband-diags-48.0-1.el8.x86_64 5/12
Verifying : libaec-1.0.2-3.el8.x86_64 6/12
Verifying : hdf5-1.10.5-4.el8.x86_64 7/12
Verifying : libjwt-1.12.1-7.el8.x86_64 8/12
Verifying : slurm-20.11.9-1.el8.x86_64 9/12
Verifying : slurm-libs-20.11.9-1.el8.x86_64 10/12
Verifying : slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
Verifying : slurm-slurmd-20.11.9-1.el8.x86_64 12/12

Installed:
  freeipmi-1.6.14-2.el8.x86_64  hdf5-1.10.5-4.el8.x86_64  infiniband-diags-48.0-1.el8.x86_64  libaec-1.0.2-3.el8.x86_64
  libjwt-1.12.1-7.el8.x86_64  mariadb-connector-c-3.1.11-2.el8_3.x86_64  mariadb-connector-c-config-3.1.11-2.el8_3.noarch  munge-0.5.13-3.el8_10.x86_64
  slurm-20.11.9-1.el8.x86_64  slurm-libs-20.11.9-1.el8.x86_64  slurm-slurmctld-20.11.9-1.el8.x86_64  slurm-slurmd-20.11.9-1.el8.x86_64

Complete!
liquitecompute@headnode ~]$
```

```
sudo dnf install -y nfs-utils munge
```

```

Transaction Summary
=====
Install 6 Packages

Total download size: 836 k
Installed size: 2.1 M
Downloading Packages:
(1/6): libveto-libev-0.3.2-2.el8.x86_64.rpm           378 kB/s | 15 kB  00:00
(2/6): libev-4.24-6.el8.x86_64.rpm                   1.0 MB/s | 51 kB  00:00
(3/6): gssproxy-0.8.0-21.el8.x86_64.rpm              1.8 MB/s | 118 kB  00:00
(4/6): keyutils-1.5.10-9.el8.x86_64.rpm              2.0 MB/s | 65 kB  00:00
(5/6): rpcbind-1.2.5-10.el8.x86_64.rpm               2.7 MB/s | 69 kB  00:00
(6/6): nfs-utils-2.3.3-68.el8_10.x86_64.rpm          6.0 MB/s | 518 kB  00:00
-----
Total                                                    451 kB/s | 836 kB  00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Running scriptlet: rpcbind-1.2.5-10.el8.x86_64      1/1
  Installing : rpcbind-1.2.5-10.el8.x86_64          1/6
  Running scriptlet: rpcbind-1.2.5-10.el8.x86_64     1/6
  Installing : keyutils-1.5.10-9.el8.x86_64         1/6
  Installing : libev-4.24-6.el8.x86_64              3/6
  Installing : libveto-libev-0.3.2-2.el8.x86_64     4/6
  Installing : gssproxy-0.8.0-21.el8.x86_64         5/6
  Running scriptlet: gssproxy-0.8.0-21.el8.x86_64    5/6
  Running scriptlet: nfs-utils-1:2.3.3-68.el8_10.x86_64 6/6
  Installing : nfs-utils-1:2.3.3-68.el8_10.x86_64   6/6
  Running scriptlet: nfs-utils-1:2.3.3-68.el8_10.x86_64 6/6
[12384.436165] RPC: Registered named UNIX socket transport module.
[12384.436383] RPC: Registered udp transport module.
[12384.436640] RPC: Registered tcp transport module.
[12384.436762] RPC: Registered tcp NFSv4.1 backchannel transport module.
  Verifying : libev-4.24-6.el8.x86_64              1/6
  Verifying : libveto-libev-0.3.2-2.el8.x86_64     2/6
  Verifying : gssproxy-0.8.0-21.el8.x86_64         3/6
  Verifying : keyutils-1.5.10-9.el8.x86_64         4/6
  Verifying : nfs-utils-1:2.3.3-68.el8_10.x86_64   5/6
  Verifying : rpcbind-1.2.5-10.el8.x86_64          6/6

Installed:
gssproxy-0.8.0-21.el8.x86_64  keyutils-1.5.10-9.el8.x86_64  libev-4.24-6.el8.x86_64  libveto-libev-0.3.2-2.el8.x86_64  nfs-utils-1:2.3.3-68.el8_10.x86_64
rpcbind-1.2.5-10.el8.x86_64

Complete!
ignitecompute@headnode ~]$

```

7. NFS Shared Storage

Head Node

```

sudo mkdir /shared
sudo chmod 777 /shared
sudo nano /etc/exports
# Add: /shared *(rw,sync,no_root_squash)
sudo systemctl enable nfs-server
sudo systemctl start nfs-server

```

```

ignitecompute@headnode ~]$ sudo systemctl status nfs-server
[sudo] password for ignitecompute:
• nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; disabled; vendor preset: disabled)
   Active: inactive (dead)
ignitecompute@headnode ~]$ sudo systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server.service.
ignitecompute@headnode ~]$ sudo systemctl status nfs-server
• nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; vendor preset: disabled)
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: inactive (dead)
ignitecompute@headnode ~]$ sudo exportfs -a
ignitecompute@headnode ~]$ sudo exportfs -v
/ shared      192.168.231.11(sync,wdelay,hide,no_subtree_check,sec=sys,rw,no_root_squash,no_all_squash)
/ shared      192.168.231.12(sync,wdelay,hide,no_subtree_check,sec=sys,rw,no_root_squash,no_all_squash)
ignitecompute@headnode ~]$

```


Compute Nodes

```
sudo mkdir /shared
sudo mount head-node:/shared /shared
df -h
```

Mounted shared directory

```
GNU nano 2.9.8 /etc/exports

/shared 192.168.231.11(rw,sync,no_root_squash)
/shared 192.168.231.12(rw,sync,no_root_squash)
```

8. MUNGE Authentication

Head Node

```
sudo create-munge-key
scp /etc/munge/munge.key compute-node-1:/etc/munge/
scp /etc/munge/munge.key compute-node-2:/etc/munge/
```

```
ignitecompute@headnode ~]$ sudo dnf install -y munge munge-libs munge-devel
[sudo] password for ignitecompute:
Last metadata expiration check: 1:28:04 ago on Tue 31 Mar 2025 10:50:48 AM EDT.
Package munge-0.5.13-3.el8_10.x86_64 is already installed.
Package munge-libs-0.5.13-3.el8_10.x86_64 is already installed.
Dependencies resolved.
=====
Package                               Architecture      Version           Repository        Size
=====
Installing:
munge-devel                            x86_64            0.5.13-3.el8_10  powertools        27 k
=====
Transaction Summary
=====
Install 1 Package
Total download size: 27 k
Installed size: 23 k
Downloading Packages:
munge-devel-0.5.13-3.el8_10.x86_64.rpm                                41 kB/s | 27 kB  00:00
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :
  Installing     : munge-devel-0.5.13-3.el8_10.x86_64                1/1
  Running scriptlet: munge-devel-0.5.13-3.el8_10.x86_64                1/1
  Verifying      : munge-devel-0.5.13-3.el8_10.x86_64                1/1
Installed:
munge-devel-0.5.13-3.el8_10.x86_64
Complete!
ignitecompute@headnode ~]$
```

All Nodes

```
sudo chown munge: /etc/munge/munge.key
sudo chmod 400 /etc/munge/munge.key
sudo systemctl enable munge
sudo systemctl start munge
munge -n | unmunge
```

9. Slurm Configuration

```
Total 1.7 MB/s | 8.3 MB 00:04
Extra Packages for Enterprise Linux 8 - x86_64 1.6 MB/s | 1.6 kB 00:00
Importing GPG key 0x2F86D6A1:
  Userid : "Fedora EPEL (8) <epel@fedoraproject.org>"
  Fingerprint: 94E2 79ED 80BF 25B2 1810 ADP1 21EA 45AB 2F86 D6A1
  From : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-8
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing :
  Installing : slurm-libs-20.11.9-1.el8.x86_64 1/1
  Installing : libaec-1.0.2-3.el8.x86_64 1/12
  Installing : hdf5-1.10.5-4.el8.x86_64 2/12
  Installing : libjwt-1.12.1-7.el8.x86_64 3/12
  Installing : infiniband-diags-48.0-1.el8.x86_64 4/12
  Installing : freeipmi-1.6.14-2.el8.x86_64 5/12
  Running scriptlet: freeipmi-1.6.14-2.el8.x86_64 6/12
  Running scriptlet: munge-0.5.13-3.el8_10.x86_64 6/12
  Installing : munge-0.5.13-3.el8_10.x86_64 7/12
  Running scriptlet: munge-0.5.13-3.el8_10.x86_64 7/12
  Installing : mariadb-connector-c-config-3.1.11-2.el8_3.noarch 8/12
  Installing : mariadb-connector-c-3.1.11-2.el8_3.x86_64 9/12
  Installing : slurm-20.11.9-1.el8.x86_64 10/12
  Installing : slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
  Running scriptlet: slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
  Installing : slurm-slurmd-20.11.9-1.el8.x86_64 12/12
  Running scriptlet: slurm-slurmd-20.11.9-1.el8.x86_64 12/12
  Verifying : mariadb-connector-c-3.1.11-2.el8_3.x86_64 1/12
  Verifying : mariadb-connector-c-config-3.1.11-2.el8_3.noarch 2/12
  Verifying : munge-0.5.13-3.el8_10.x86_64 3/12
  Verifying : freeipmi-1.6.14-2.el8.x86_64 4/12
  Verifying : infiniband-diags-48.0-1.el8.x86_64 5/12
  Verifying : libaec-1.0.2-3.el8.x86_64 6/12
  Verifying : hdf5-1.10.5-4.el8.x86_64 7/12
  Verifying : libjwt-1.12.1-7.el8.x86_64 8/12
  Verifying : slurm-20.11.9-1.el8.x86_64 9/12
  Verifying : slurm-libs-20.11.9-1.el8.x86_64 10/12
  Verifying : slurm-slurmctld-20.11.9-1.el8.x86_64 11/12
  Verifying : slurm-slurmd-20.11.9-1.el8.x86_64 12/12

Installed:
  freeipmi-1.6.14-2.el8.x86_64  hdf5-1.10.5-4.el8.x86_64  infiniband-diags-48.0-1.el8.x86_64  libaec-1.0.2-3.el8.x86_64  munge-0.5.13-3.el8_10.x86_64
  libjwt-1.12.1-7.el8.x86_64  mariadb-connector-c-3.1.11-2.el8_3.x86_64  mariadb-connector-c-config-3.1.11-2.el8_3.noarch  slurm-20.11.9-1.el8.x86_64  slurm-slurmctld-20.11.9-1.el8.x86_64
  slurm-20.11.9-1.el8.x86_64  slurm-libs-20.11.9-1.el8.x86_64  slurm-slurmd-20.11.9-1.el8.x86_64

Complete!
liquitecompute@headnode ~]$
```

Copy to Compute Nodes

```
scp /etc/slurm/slurm.conf compute-node-1:/etc/slurm/
scp /etc/slurm/slurm.conf compute-node-2:/etc/slurm/
```

Start Services

Head Node:

```
sudo systemctl enable slurmctld
sudo systemctl start slurmctld
```

Compute Nodes:

```
sudo systemctl enable slurmd
sudo systemctl start slurmd
```

Benchmarks

a) HPL:

```
[head-node@head-node ~]$ mpirun -np 1 ./xhpl
-----
mpirun was unable to launch the specified application as it could not access
or execute an executable:

Executable: ./xhpl
Node: head-node

while attempting to start process rank 0.
-----
[head-node@head-node ~]$
```

b) LAMMPS:

```
Installed:
  git-2.43.7-1.el8_10.x86_64      git-core-2.43.7-1.el8_10.x86_64      git-core-doc-2.43.7-1.el8_10.noarch
  perl-Git-2.43.7-1.el8_10.noarch  perl-TermReadKey-2.37-7.el8.x86_64

Complete!
[head-node@head-node ~]$ git clone https://github.com/lammps/lammps.git
Cloning into 'lammps'...
remote: Enumerating objects: 458638, done.
remote: Counting objects: 100% (1117/1117), done.
remote: Compressing objects: 100% (315/315), done.
remote: Total 458638 (delta 975), reused 820 (delta 802), pack-reused 457521 (from 3)
Receiving objects: 100% (458638/458638), 835.16 MiB | 3.06 MiB/s, done.
Resolving deltas: 100% (376405/376405), done.
Updating files: 100% (13283/13283), done.
[head-node@head-node ~]$ cd lammps
[head-node@head-node lammps]$ mkdir build
[head-node@head-node lammps]$ cd build
[head-node@head-node build]$ cmake ../cmake -D BUILD_MPI=on
-- The CXX compiler identification is GNU 8.5.0
-- The C compiler identification is GNU 8.5.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found Git: /usr/bin/git (found version "2.43.7")
-- Running check for auto-generated files from make-based build system
CMake Warning at CMakeLists.txt:183 (message):
  Using the GNU compilers version 8.5.0 with C++17 or later is not
  recommended. Please use GNU compilers version 9.3 or later
```

```
[head-node@head-node ~]$ sudo dnf install gcc gcc-c++ make cmake openmpi openmpi-devel fftw-devel
Last metadata expiration check: 3:31:13 ago on Tue 31 Mar 2026 10:47:25 AM EDT.
Package gcc-8.5.0-28.el8_10.x86_64 is already installed.
Package make-1:4.2.1-11.el8.x86_64 is already installed.
Package openmpi-2:4.1.1-5.el8.x86_64 is already installed.
Package openmpi-devel-2:4.1.1-5.el8.x86_64 is already installed.
Dependencies resolved.
```

Package	Architecture	Version
Installing:		
cmake	x86_64	3.26.5-2.el8
fftw-devel	x86_64	3.3.5-11.el8
gcc-c++	x86_64	8.5.0-28.el8_10
Installing dependencies:		
cmake-data	noarch	3.26.5-2.el8
cmake-filesystem	x86_64	3.26.5-2.el8
cmake-rpm-macros	noarch	3.26.5-2.el8
emacs-filesystem	noarch	1:26.1-15.el8_10
fftw	x86_64	3.3.5-11.el8
fftw-libs	x86_64	3.3.5-11.el8
fftw-libs-double	x86_64	3.3.5-11.el8
fftw-libs-long	x86_64	3.3.5-11.el8
fftw-libs-quad	x86_64	3.3.5-11.el8
fftw-libs-single	x86_64	3.3.5-11.el8
libstdc++-devel	x86_64	8.5.0-28.el8_10
libuv	x86_64	1:1.41.1-2.el8_10
vim-filesystem	noarch	2:8.0.1763-22.el8_10

Transaction Summary

Install 16 Packages

Total download size: 34 M

Installed size: 97 M

Is this ok [y/N]: y

```

Create atoms CPU = 0.005 seconds
Generated 0 of 0 mixed pair_coeff terms from geometric mixing rule
Neighbor list info ...
  update: every = 20 steps, delay = 0 steps, check = no
  max neighbors/atom: 2000, page size: 100000
  master list distance cutoff = 2.8
  ghost atom cutoff = 2.8
  binsize = 1.4, bins = 24 24 24
  1 neighbor lists, perpetual/occasional/extra = 1 0 0
  (1) pair lj/cut, perpetual
    attributes: half, newton on
    pair build: half/bin/atomonly/newton
    stencil: half/bin/3d
    bin: standard
Setting up Verlet run ...
Unit style      : lj
Current step    : 0
Time step       : 0.005
Per MPI rank memory allocation (min/avg/max) = 13.82 | 13.82 | 13.82 Mbytes

```

Step	Temp	E_pair	E_mol	TotEng	Press
0	1.44	-6.7733681	0	-4.6134356	-5.0197073
100	0.7574531	-5.7585055	0	-4.6223613	0.20726105

```

Loop time of 1.53494 on 1 procs for 100 steps with 32000 atoms

Performance: 28144.398 tau/day, 65.149 timesteps/s, 2.005 Matom-step/s
98.5% CPU use with 1 MPI tasks x 1 OpenMP threads

MPI task timing breakdown:

```

Section	min time	avg time	max time	%varavg	%total
Pair	1.2076	1.2076	1.2076	0.0	83.89
Neigh	0.20281	0.20281	0.20281	0.0	13.21
Comm	0.01305	0.01305	0.01305	0.0	0.85
Output	0.00010517	0.00010517	0.00010517	0.0	0.01
Modify	0.026158	0.026158	0.026158	0.0	1.70
Other		0.005203			0.34

```

Nlocal:          32000 ave          32000 max          32000 min
Histogram: 1 0 0 0 0 0 0 0 0
Nghost:          19657 ave          19657 max          19657 min
Histogram: 1 0 0 0 0 0 0 0 0
Neighs:    1.20283e+06 ave 1.20283e+06 max 1.20283e+06 min
Histogram: 1 0 0 0 0 0 0 0 0

Total # of neighbors = 1202833
Ave neighs/atom = 37.588531
Neighbor list builds = 5
Dangerous builds not checked
Total wall time: 0:00:01
[head-node@head-node build]$

```

HPCC

```

Is this ok [y/N]: y
Downloading Packages:
(1/16): hicolor-icon-theme-0.17-2.el8.noarch.rpm
(2/16): jbigkit-libs-2.1-14.el8.x86_64.rpm
(3/16): libXpm-3.5.12-11.el8.x86_64.rpm
(4/16): gd-2.2.5-7.el8.x86_64.rpm
(5/16): libjpeg-turbo-1.5.3-14.el8_10.x86_64.rpm
(6/16): libtiff-4.0.9-36.el8_10.x86_64.rpm
(7/16): libwebp-1.0.0-11.el8_10.x86_64.rpm
(8/16): php-gd-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(9/16): php-json-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(10/16): php-pdo-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(11/16): php-process-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(12/16): php-common-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(13/16): php-xml-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(14/16): libxslt-1.1.32-6.3.el8_10.x86_64.rpm
(15/16): php-cli-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64.rpm
(16/16): phoronix-test-suite-10.8.4-2.el8.noarch.rpm
-----

```

Total

Extra Packages for Enterprise Linux 8 - x86_64

Importing GPG key 0x2F86D6A1:

Userid : "Fedora EPEL (8) <epel@fedoraproject.org>"

Fingerprint: 94E2 79EB 8D8F 25B2 1810 ADF1 21EA 45AB 2F86 D6A1

From : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-8

Is this ok [y/N]:

```

Installing : phoronix-test-suite-10.8.4-2.el8.noarch
Installed:
  gd-2.2.5-7.el8.x86_64
  jbigkit-libs-2.1-14.el8.x86_64
  libjpeg-turbo-1.5.3-14.el8_10.x86_64
  libwebp-1.0.0-11.el8_10.x86_64
  phoronix-test-suite-10.8.4-2.el8.noarch
  php-common-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  php-json-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  php-process-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  hicolor-icon-theme-0.17-2.el8.noarch
  libXpm-3.5.12-11.el8.x86_64
  libtiff-4.0.9-36.el8_10.x86_64
  libxslt-1.1.32-6.3.el8_10.x86_64
  php-cli-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  php-gd-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  php-pdo-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64
  php-xml-7.2.24-1.module+el8.4.0+413+c9202dda.x86_64

```

Complete!

[head-node@head-node ~]\$

```
Nlocal:      32000 ave      32000 max      32000 min
Histogram: 1 0 0 0 0 0 0 0 0
Nghost:      19657 ave      19657 max      19657 min
Histogram: 1 0 0 0 0 0 0 0 0
Neighs:      1.20283e+06 ave 1.20283e+06 max 1.20283e+06 min
Histogram: 1 0 0 0 0 0 0 0 0
```

```
Total # of neighbors = 1202833
Ave neighs/atom = 37.588531
Neighbor list builds = 5
Dangerous builds not checked
Total wall time: 0:00:01
[head-node@head-node build1]$ cd
[head-node@head-node ~]$ sudo dnf install phoronix-test-suite
[sudo] password for head-node:
Last metadata expiration check: 4:29:09 ago on Tue 31 Mar 2026 10:47:25 AM EDT.
Dependencies resolved.
```

Package	Architecture	Version
Installing:		
phoronix-test-suite	noarch	10.8.4-2.el8
Installing dependencies:		
gd	x86_64	2.2.5-7.el8
hicolor-icon-theme	noarch	0.17-2.el8
jbigkit-libs	x86_64	2.1-14.el8
libXpm	x86_64	3.5.12-11.el8
libjpeg-turbo	x86_64	1.5.3-14.el8_10
libtiff	x86_64	4.0.9-36.el8_10
libwebp	x86_64	1.0.0-11.el8_10
libxslt	x86_64	1.1.32-6.3.el8_10
php-cli	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-common	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-gd	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-json	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-pdo	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-process	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
php-xml	x86_64	7.2.24-1.module+el8.4.0+413+c9202dda
Enabling module streams:		
httpd		2.4
nginx		1.14
php		7.2

Transaction Summary

Install 16 Packages

Total download size: 9.4 M
Installed size: 36 M
Is this ok [y/N]:

The following dependencies are needed and will be installed:

```
- blas
- blas-devel
- openblas-devel
- lapack-devel
```

This process may take several minutes.

Error: Unable to find a match: blas-devel openblas-devel lapack-devel

Last metadata expiration check: 5:07:02 ago on Tue 31 Mar 2026 10:47:25 AM EDT.

Package blas-3.8.0-8.el8.x86_64 is already installed.

No match for argument: blas-devel

No match for argument: openblas-devel

No match for argument: lapack-devel

Phoronix Test Suite v10.8.4

To Install: pts/hpcc-1.2.8

Determining File Requirements

Searching Download Caches

1 Test To Install

9MB Of Disk Space Is Needed

12 Seconds Estimated Install Time

pts/hpcc-1.2.8:

Test Installation 1 of 1

1 File Needed [0.63 MB / 1 Minute]

File Found: hpcc-1.5.0.tar.gz

Approximate Install Size: 9 MB

Estimated Install Time: 12 Seconds

Installing Test @ 15:54:28

[NOTICE] Supported install-time optional variables include \$MPI_PATH, \$MPI_INCLUDE, \$MPI_CC, \$MPI_LIBS, \$LA_PATH, \$LD_FLAGS, and \$MPI_LD

PennyLane

```
Collecting toml
  Downloading toml-0.10.2-py2.py3-none-any.whl (16 kB)
Collecting appdirs
  Downloading appdirs-1.4.4-py2.py3-none-any.whl (9.6 kB)
Collecting autograd
  Downloading autograd-1.6.2-py3-none-any.whl (49 kB)
INFO: pip is looking at multiple versions of pennylane-lightning to determine which version is compatible with other requirements. This could take a while.
Collecting pennylane-lightning
  Downloading PennyLane_Lightning-0.36.0-py3-none-any.whl (121 kB)
  Downloading PennyLane_Lightning-0.35.1-py3-none-any.whl (103 kB)
  Downloading PennyLane_Lightning-0.35.0-py3-none-any.whl (103 kB)
  Downloading PennyLane_Lightning-0.34.0-py3-none-any.whl (99 kB)
  Downloading PennyLane_Lightning-0.33.1-py3-none-any.whl (92 kB)
  Downloading PennyLane_Lightning-0.33.0-py3-none-any.whl (92 kB)
  Downloading PennyLane_Lightning-0.32.0-py3-none-any.whl (84 kB)
INFO: pip is looking at multiple versions of pennylane-lightning to determine which version is compatible with other requirements. This could take a while.
  Downloading PennyLane_Lightning-0.31.0-py3-none-any.whl (236 kB)
  Downloading PennyLane_Lightning-0.30.0-py3-none-any.whl (236 kB)
Collecting future>=0.15.2
  Downloading future-1.0.0-py3-none-any.whl (491 kB)
Collecting decorator<5,>=4.3
  Downloading decorator-4.4.2-py2.py3-none-any.whl (9.2 kB)
Collecting charset-normalizer<=2.0.8
  Downloading charset-normalizer-2.0.12-py3-none-any.whl (39 kB)
Collecting idna<4,>=2.5
  Downloading idna-3.10-py3-none-any.whl (70 kB)
Collecting urllib3<1.27,>=1.21.1
  Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
Collecting certifi>=2017.4.17
  Downloading certifi-2025.4.26-py3-none-any.whl (159 kB)
Installing collected packages: urllib3, numpy, idna, future, decorator, charset-normalizer, certifi, toml, semantic-version, scipy, networkx, requests, networkx, cachetools, autoray, autograd, appdirs, pennylane-lightning, pennylane
Successfully installed appdirs-1.4.4 autograd-1.6.2 autoray-0.6.3 cachetools-4.2.4 certifi-2025.4.26 charset-normalizer-2.0.12 decorator-4.4.2 future-1.0.0 idna-3.10 networkx-2.5.1 numpy-1.19.5 pennylane-0.29.1 pennylane-lightning-0.30.0 requests-2.27.1 networkx-0.11.0 scipy-1.5.4 semantic-version-2.10.0 toml-0.10.2 ur
(pennylane env) [ignitecompute@headnode ~]$
```

MILC


```

Lignitecompute@headnode ~]$ git clone https://github.com/milc-qcd/milc-qcd.git
Cloning into 'milc-qcd'...
remote: Enumerating objects: 34334, done.
remote: Counting objects: 100% (2217/2217), done.
remote: Compressing objects: 100% (288/288), done.
remote: Total 34334 (delta 2182), reused 1929 (delta 1929), pack-reused 32117 (from 4)
Receiving objects: 100% (34334/34334), 151.23 MiB | 5.62 MiB/s, done.
Resolving deltas: 100% (25277/25277), done.
Lignitecompute@headnode ~]$ cd milc-qcd
Lignitecompute@headnode milc-qcd]$ ls
arb_dirac_eigen      diffn3.pl           getints.sh          ks_imp_rhmc         Make_test_template  rcorr
arb_dirac_invert     diffn.pl            getsizeof.sh        ks_imp_utilities    Make_time_template  README
arb_overlap          doc                 gluon_prop          ks_measure           Make_time_work      README_CMAKE.md
binary_samples       ext_src             Grid                ks_spectrum         Make_unpack          README.md
cc.pl               file_combine        groom               libraries            NEWS                 README_RELEASE_NOTES
check_malloc.pl      file_utilities      h_dibaryon          LICENSE              propagating_form_factor
check.pl            gauge_utilities     headtail.pl         ludmila              pure_gauge           README_UNPACK
clover_dynamical     generic             heavy               Make_contents       pure_gauge_old       schroed_cl_inv
clover_hybrids       generic_clover      hget_heavy_to_light Make_file             pw_nr_meson          schroed_ks_dyn
clover_invert        generic_form        hvj_qpot            Make_sample_template python2lib            schroed_pg
clover_invert2       generic_ks          include             Make_tar             python3lib            scidac
cmake               generic_pg          ks_dynamical        Make_template_combos qcdml                seterrfile.pl
CMakeLists.txt       generic_schroed     ks_eigen             Make_template_gop    QPhiX_JLab           seterr.sh
compare.sh           generic_u1          ks_hl_spectrum      Make_template_scidac QPhiX_MILC           short_tag.sh
dense_static_su3     generic_wilson      ks_imp_dyn          Make_test_all        quda                  smooth_inst
Lignitecompute@headnode milc-qcd]$ ls milc
Lignitecompute@headnode ~]$ mpirun -np 4 milc test.input
-----
There are not enough slots available in the system to satisfy the 4
slots that were requested by the application:

    milc

Either request fewer slots for your application, or make more slots
available for use.

A "slot" is the Open MPI term for an allocatable unit where we can
launch a process. The number of slots available are defined by the
environment in which Open MPI processes are run:

1. Hostfile, via "slots=N" clauses (N defaults to number of
   processor cores if not provided)
2. The --host command line parameter, via a ":N" suffix on the
   hostname (N defaults to 1 if not provided)
3. Resource manager (e.g., SLURM, PBS/Torque, LSF, etc.)
4. If none of a hostfile, the --host command line parameter, or an
   RM is present, Open MPI defaults to the number of processor cores

In all the above cases, if you want Open MPI to default to the number
of hardware threads instead of the number of processor cores, use the
--use-hwthread-cpus option.

Alternatively, you can use the --oversubscribe option to ignore the
number of available slots when deciding the number of processes to
launch.
-----
Lignitecompute@headnode ~]$

```

Optimizing a Matlab Script

```

GNU nano 2.9.8                                     matlab_job.sh

#!/bin/bash
#SBATCH --job-name=matlab_parfor
#SBATCH --nodes=1
#SBATCH --ntasks=8
#SBATCH --time=01:00:00
#SBATCH --partition=compute

module load matlab

srun matlab -nodisplay -nosplash -r "myScript; exit"

```

```

GNU nano 2.9.8                                     myScript.m

parpool('local',8);
parfor i = 1:N
    results(i) = heavyComputation(i);
end
save('results.mat', 'results');
exit

```

Testing Connectivity

```

[computer-node2@computernode2 ~]$ ping 192.168.231.10
PING 192.168.231.10 (192.168.231.10) 56(84) bytes of data.
64 bytes from 192.168.231.10: icmp_seq=1 ttl=64 time=0.637 ms
64 bytes from 192.168.231.10: icmp_seq=2 ttl=64 time=0.927 ms
64 bytes from 192.168.231.10: icmp_seq=3 ttl=64 time=0.909 ms
64 bytes from 192.168.231.10: icmp_seq=4 ttl=64 time=0.929 ms
64 bytes from 192.168.231.10: icmp_seq=5 ttl=64 time=0.803 ms
64 bytes from 192.168.231.10: icmp_seq=6 ttl=64 time=0.922 ms
64 bytes from 192.168.231.10: icmp_seq=7 ttl=64 time=0.948 ms
64 bytes from 192.168.231.10: icmp_seq=8 ttl=64 time=0.976 ms
64 bytes from 192.168.231.10: icmp_seq=9 ttl=64 time=0.946 ms
^C
--- 192.168.231.10 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8077ms
rtt min/avg/max/mdev = 0.637/0.888/0.976/0.104 ms
[computer-node2@computernode2 ~]$ ping 192.168.231.11
PING 192.168.231.11 (192.168.231.11) 56(84) bytes of data.
64 bytes from 192.168.231.11: icmp_seq=1 ttl=64 time=0.800 ms
64 bytes from 192.168.231.11: icmp_seq=2 ttl=64 time=0.958 ms
64 bytes from 192.168.231.11: icmp_seq=3 ttl=64 time=0.917 ms
64 bytes from 192.168.231.11: icmp_seq=4 ttl=64 time=0.896 ms
64 bytes from 192.168.231.11: icmp_seq=5 ttl=64 time=0.947 ms
64 bytes from 192.168.231.11: icmp_seq=6 ttl=64 time=0.301 ms
64 bytes from 192.168.231.11: icmp_seq=7 ttl=64 time=0.666 ms
^C
--- 192.168.231.11 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6111ms
rtt min/avg/max/mdev = 0.301/0.783/0.958/0.220 ms
[computer-node2@computernode2 ~]$

```

```

lignitecompute@headnode ~]$ ping 192.168.231.12
PING 192.168.231.12 (192.168.231.12) 56(84) bytes of data.
64 bytes from 192.168.231.12: icmp_seq=1 ttl=64 time=0.605 ms
64 bytes from 192.168.231.12: icmp_seq=2 ttl=64 time=2.60 ms
64 bytes from 192.168.231.12: icmp_seq=3 ttl=64 time=0.931 ms
64 bytes from 192.168.231.12: icmp_seq=4 ttl=64 time=2.26 ms
64 bytes from 192.168.231.12: icmp_seq=5 ttl=64 time=0.919 ms
64 bytes from 192.168.231.12: icmp_seq=6 ttl=64 time=0.933 ms
64 bytes from 192.168.231.12: icmp_seq=7 ttl=64 time=0.925 ms
64 bytes from 192.168.231.12: icmp_seq=8 ttl=64 time=0.880 ms
^C
--- 192.168.231.12 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7024ms
rtt min/avg/max/mdev = 0.605/1.257/2.603/0.692 ms
lignitecompute@headnode ~]$ ping 192.168.231.11
PING 192.168.231.11 (192.168.231.11) 56(84) bytes of data.
64 bytes from 192.168.231.11: icmp_seq=1 ttl=64 time=0.528 ms
64 bytes from 192.168.231.11: icmp_seq=2 ttl=64 time=0.936 ms
64 bytes from 192.168.231.11: icmp_seq=3 ttl=64 time=0.956 ms
64 bytes from 192.168.231.11: icmp_seq=4 ttl=64 time=0.957 ms
64 bytes from 192.168.231.11: icmp_seq=5 ttl=64 time=0.922 ms
64 bytes from 192.168.231.11: icmp_seq=6 ttl=64 time=0.902 ms
64 bytes from 192.168.231.11: icmp_seq=7 ttl=64 time=0.958 ms
^C
--- 192.168.231.11 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6069ms
rtt min/avg/max/mdev = 0.528/0.879/0.958/0.150 ms
lignitecompute@headnode ~]$

```

12. Troubleshooting

Simulate failure:

```
sudo systemctl stop slurmd
```

Fix:

```
sudo systemctl start slurmd
```

IP Addresses failure:

```
sudo nmcli device connect enp0s3
```

```
sudo nmcli device connect enp0s8
```

checking DNS:

```
ping -c 4 google.com
```

fix DNS:

```
sudo nano /etc/resolv.config
```

Add

```
Name server 8.8.8.8
```

13. Security Measures

- SSH key authentication
- Limited sudo access
- Firewall/SELinux configuration

14. Reproducibility

- Store scripts and configs in /shared
- Document versions of software

15. Conclusion

The HPC cluster was successfully implemented, benchmarked, and tested. Performance improved as expected with multiple nodes, demonstrating the effectiveness of parallel computing.
